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ENGINEERING STATISTICS.

Statistics can be defined as the measurement of uncertainty.

Statistics is a discipline that allows researchers to evaluate conclusions derived from sample data. In practice, it refers to a scientific approach used to

* Collect data
* Interpret and analyse data
* Assess the reliability of conclusions based on sample data.

A statistic is a characteristic of a sample. Generally, a statistic is used to estimate the value of a population parameter.

Statistics is the art and science of collecting, classifying, presenting, interpreting and analysing numerical data, as well as making conclusions about the system from which the data was obtained. The field of statistics can be broken into 2 major areas: descriptive statistics and inferential statistics

Statistics is collection, analysis and use of data to understand and predict factors of interest in a scientific way to make decisions, solve problems and design products or processes

Why learn statistics.

* Potential outcome of things can be predicted with help of statistics, where other techniques are not adequate
* Helps in decision making based on what is already known (climate disasters, agricultural production etc)

Role of statistics in Engg.

Problems of interest are solved by Engineers using scientific method.

1. Describe problem.
2. Identify affecting factors.

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3. Propose a model
4. Collect data from experiments to validate model.
5. Refine the model based on data.
6. Develop solution.
7. Collect data to confirm solution effectiveness & efficiency.
8. Conclude/ give recommendations.

Role of stats.

* Collection, presentation, analysis and use of data to make decisions.
* Help describe and understand variability.
* Make conclusions about a population based on sample observations (statistical inference)
  eg if sample data exists is it enough to infer about popln?
* Distinguish between inherent variability & changes due to modification of i/p.

Applications

* Process improvement (Mean productivity of processes etc)
* Product design (effect of parameters)
* Health care Eg Is a new vaccination program going to be effective.
* Strategic decision making (decision based on surveys)
* Prediction of parameters (Financial returns, inflation, weather etc)

Statistics < Descriptive
Inferential statistics.

Descriptive statistics is the branch of statistics which characterizes and summarizes the most prominent features of a given set of data. Some of the commonly used descriptive measures include means, medians, std. devns, percentiles, graphs, tables and charts. Descriptive stats can be used to describe the elements of a popln as a whole, or they can be used to describe data that represent

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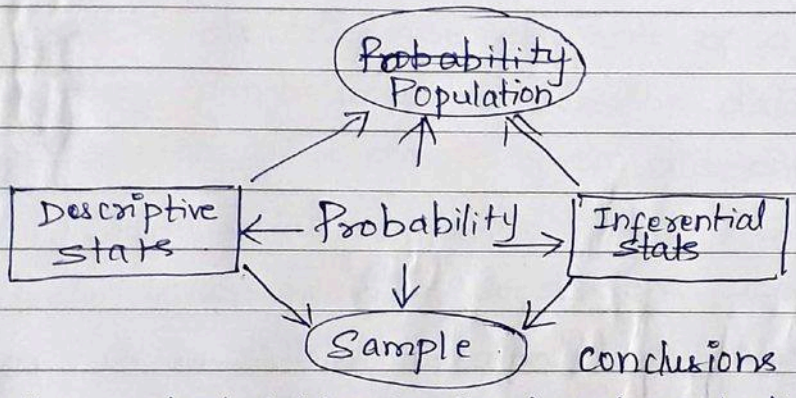
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just a sample of elements from the entire population.

Inferential statistics is the branch of statistics that deals with drawing conclusions about a population based on information obtained from a sample drawn from that population.

Probability is the bridge b/w descriptive stats & inferential stats. It is also a link b/w a population and a sample. When certain properties of a population are known or can be assumed, the probability principles involved allow one to deduce expected properties of a sample.



Alternatively, prob, together with sample data allows one to evaluate the amount of uncertainty when making conclusions about a population.

Hence probability is the heart of both deductive and inferential processes involved in statistical analysis.

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* Numerical data arising in any sphere of human experiences
  → Eg: Birth-death data
  → Crime data — food products consumption
  → agricultural production of different crops
  → industrial production in various sectors (heavy eqpt, medium, small scale industry)
  → earnings thro export of various commodities.
  → runs scored / wickets taken / goals scored / matches won / running times
  → Social → age at marriage / age of parent at birth of child / infant mortality rate
  → medicine → incidence of certain diseases / control / survival rates / effects of medicines / side effects / relation b/w certain diseases and habits.
  → Accidents → road, rail, air, ship, fatality rate due to these.
  → number of vehicles, length of roads/rails, number of valid licence holders.
  → no. of votes, no. of constituencies etc.
  → planets / solar system / stars / galaxies / comets / astronomical events such as solar and lunar eclipse, positions of certain stars and planets → (cosmology)
  → chemicals and their compositions, electrons, neutrons.
  → genetic studies → human behavior
  → metallurgical study.
  → biometric studies
  → % of defective items, no of items produced
  → consumers of certain products their capacity to spend, their income levels & profession.
  → Weather data.
  → placement data

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History

* The word "statistics" was coined by the German scholar Gottfried Achenwall around middle of 18th century.
* Used for the first time in Britain by John Sinclair (1791-1799)
* Statistics derived from "state"
* Appeared in Encyclopedia Britannica in 1797
* Earliest work on statistical methodology was related to normal distn (De Moivre 1733. the paper discovered by Karl pearson in 1924)
* French mathematician Laplace in his manuscript in 1783, said that normal curve can be used for describing the prob. distn of errors.
* (1809) Gauss developed normal distn → in a study of theory of motions of heavenly bodies. ↘ Gaussian distn
* Belgium mathematician, Quetlet (1796 to 1874) → usual statistics such as heights of a large no. of a people.
* Gregor Mendel, experimented did various genetic experiments & mostly on peas ~~etcetera~~ & concluded that laws of heredity are statistical in nature
* Francis Galton → hereditary genius 1869 & natural inheritance in 1889 → hereditary laws are statistical in nature & ∴ he introduced the concepts of correlation & regression
* Karl Pearson → introduced concept of statistics in systematic way. In 1901 he started famous journal called Biometrica (top journals for study of statistics) → First to talk about Chi-square test statistics → testing data for normal distn or by a gamma distn → Test of goodness of fite.
* G U Yule in 1910 gave distinction b/w population & a random sample. While R.A Fisher also gave a clear cut distinction b/w popln & samples.
* Defn of Random sampling – John Venn & C.S. Pierce
* First table of random digits was printed by Tippett in 1927
* Gosset → contributed 't' distribution → student's
* R.A Fisher → father of modern statistics → theory of foundations of statistical inference → maximum likelihood estimation, unbiased estimation, analysis of variance.

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* Neyman & E.S. Pearson → started developing testing of hypothesis
* Fishes & Neyman → credited with giving confidence inte-als → Neyman → development of stratified random sampling
* In Russia around 1939 Kolmogorov develop rigorous theory of prob. & stochastic processes

Indian History

* Kautilya's Arthashastra which was written in 321 BC has given detailed description of how to collect data on various aspects of the state
* Modern statistical methodology in India got a initiati by P.C. Mahalanobis → Studied in England & came back to India regularly use statistical tools. With his efforts India Statistical Institute was started in 1931.
* R.C. Bose contributed to the subject design of experi-ents.
* S.N. Roy, C.R. Rao contributed greatly to multivariate analysis, doe, linear models etc.
* With the efforts of Mahalanobis, central statistical orgn was established in 1951 by Govt. of India to keep, to coll the data on various aspects of state. In 1950 National Sample survey orgn. was established to collect data on various socio-economic aspects of life.

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Collecting Engg. data.

* Retrospective study using historical data.
* An observational study.
* A designed experiment.

A retrospective study would use either all or a sample of the historical process data archived over some period of time. This study may involve a lot of data, but that data may contain relatively little useful information about the problem. Furthermore some of relevant data may be missing, there may be transcription or recording errors resulting in outliers. As a result of these types of issues, statistical analysis of historical data sometimes identify interesting phenomena, but solid & reliable explanations of these phenomena are often difficult to obtain.

Observational study - Engineer observes the process or population disturbing it as little as possible & records the quantities of interest.

Designed Experiment → The engineer makes deliberate or purposeful changes in the controllable variables of the system or process, observes the resulting system output data, & then makes an inference or decision about which variables are responsible for the observed changes in output performance.

Models — A mathematical model can be defined as a description of a system using mathematical concepts and languages to facilitate proper explanation of a system or to study the effects of different components and to make predictions on patterns of behavior.

Mechanistic model → built from underlying knowledge of the basic physical mechanism that relate these variables. It is description of a complex system based on description of its components and interconnections. They mostly have a tangible, physical aspect.

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Empirical model → uses our Engg. & scientific knowledge of the phenomenon, but it is not directly developed from theoretical or first principles of underlying mechanism.

In general, one of empirical model is regression -

To estimate parameters → method of least squares that traces its origin to work by Karl Gauss.

Probability models help quantify the risks involved in statistical inference, i.e. the risks involved in decisions made every day.

Example Prediction of tides → Empirical (based on past data) w/o knowing the mechanisms.
→ Mechanistic (based on interaction of earth, Sun, moon)

Data → Raw (as collected)
→ Processed (processed after collecting)

	Raw	Processed
*	Height of all students in a particular class.	* Average height of each class of IV Sem
*	Online measurement of weight of each pack of a food item.	* Report containing avg. weight of pack for food item/day.
*	wt of a casting part measured at different manufacturing locns in diff units of measurement	* wt of components converted to a single unit for comparison purposes * Inflation rates for past 10 years
*	Down time in days of m/c in factory/shop floor	* Yearly sales figures of automobile manufacturers in India.
*	Length of phone calls for 200 persons calling thro a particular exchange	

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Data — Numerical — Discrete
                  — Continuous
     — Categorical — Nominal (Yes No)
                   — Ordinal (Grade S, A, B, C,)
                   — ~~Attribute~~ Interval (

Data types.

Categorical — Nominal
            — Ordinal

Numerical — Interval
          — Ratio.

Categorial data represents characteristics. ∴ it can represent things like a person's gender, language etc. It can also take on numerical values (Eg 1 for female & 0 for male) Note that those numbers don't have mathematical meaning.

Nominal Data – represent discrete units and are used to label variables, that have no quantitative value. It has no order.

Gender — Male
       — Female
       ~ Dichotomous → only 2 categories.

Language you speak - French
                   English.
                   German
                   Spanish.

Ordinal values represent discrete & ordered units. Hence same as nominal data except ordering matters.

Educational Background
1. Elementary
2. High School.
3. UG
4. PG

Order imp but diff b/w values is not really known.

Numerical Discrete data → Can't be measured but it can be counted.

No of heads in 100 coin flips.

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Continuous Data → represents measurements & therefore their values can't be counted but can be measured.
Ht of a person.

Interval Data → represents ordered units that have same diff.
Temperature? -10, -5, 0, +5, +10.
Problem with interval values is that they dont have a true zero. With this, we can add and subtract but cannot multiply, divide or calculate ratios.

Ratio Data → also ordered units that have same diff.
Same as interval values, with the diff that they do have an absolute zero.
Length inch — 0, 5, 10, 15.

Nominal ~~stata~~ data colln → Frequency, Proportion, Percentage
Visualization. Pie chart Bar chart.
Ordinal — Pie, bar, also use percentiles, median, mode, interquartile range.

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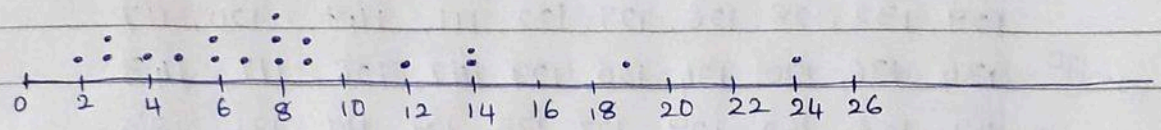
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Graphical presentation of data

Dot plot (univariate)

When only one variable is considered and plotted, dot plots are used. They are simple attractive display that spread extremes centering and voids or gaps in the data.

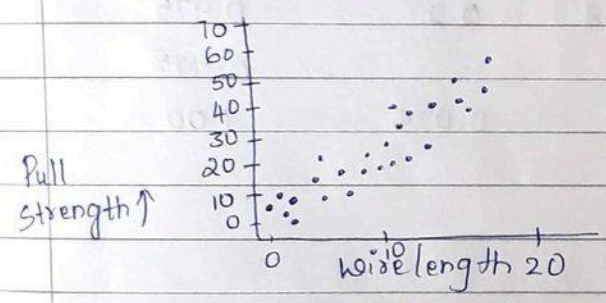
If data points are 12, 14, 9, 8, 7, 8, 4, 5, 14, 19, 2, 3, 6, 6, 5, 8, 9, 24



These are used when the number of units is relatively small say $n < 30$, or where there are relatively few distinct values represented in the data set, dot plots are effective displays.

Scatter plot — When we wish to jointly display results for 2 variable, the bivariate equivalent of dot plot is called the scatter plot.

III^{ly} a 3 dimensional scatter plot with 3 variables



Scatter plots show the region of the plane where data points fall, as well as data density associated with this region, they also suggest possible association between the variables. These can be used for large data sets also

Frequency Distribution and Histogram.

A frequency distribution is a more useful summary of data. To construct a frequency distribution, we must divide the range of data into intervals, which are usually called class intervals. If possible, the CI should be of equal width, to enhance the visualization. The no of class intervals used depends on the number of observations and the amount of scatter or dispersion in the data. Generally 5 to 20 class intervals is found satisfactory in most cases. & no of CI should increase with n. Chosing a no. of CI approx. equal to square root of the numbers of observations often works well.

For $n \leq 100 \rightarrow$ # classes $\simeq \sqrt{n}$

for $n > 100 \rightarrow$ # classes $\simeq 1 + 3.322 \log n$ ⇒ Sturges formula

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Problem 1.

The shelf life of a high speed photographic film is being investigated by the manufactures. The following data are available.

Life (days)	Life (days)	Life (days)	Life (days)
126,	131,	116,	125,
134,	120,	125,	150,
130,	149		
129,	132,	128,	126,
127,	122,	111,	148,
120,	117		
134,	136,	130,	134,
120,	129,	147,	126,
117,	143		
141,	145,	162,	129,
127,	133,	129,	140,
131,	133		

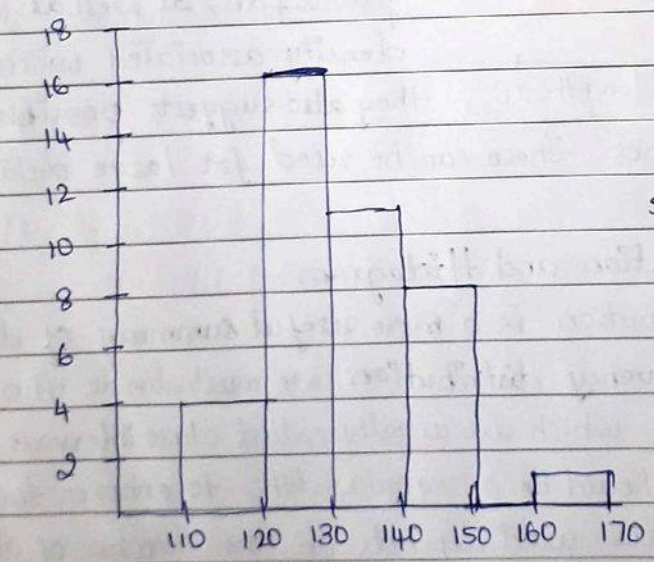
110-170

Min - 111
Max - 162

Construct the histogram & comment on the properties of the data.

C.I	Observation	Frequency	Relative frequency	Cum. relative frequency
110-120	1111	4	0.1	0.1
120-130	1111 1111 1111	16	0.4	0.5
130-140	1111 1111 1	11	0.275	0.775
140-150	1111 111	08	0.2	0.975
150-160				0.975
160-170	1	1	0.025	1.00

110 < x < 120



$$S^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1} = \frac{S_{xx}}{n-1}$$

Mean = $\frac{\bar{x}}{n}$ = 131.30.

Median - 129.5

Mode - 129.

S^2 = 113.85 S = 10.67

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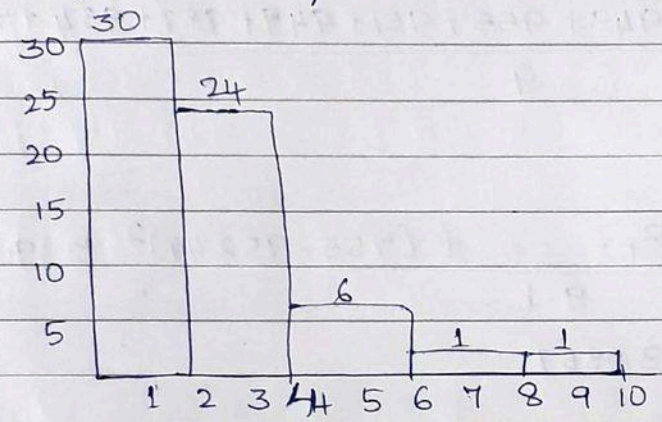
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2. 62 parts were analysed for the no of defects with the foll distribution:

No of defects i	2	3	4	5	6	7	10
No of parts with i defects	30	15	9	4	2	1	1

i) Plot a histogram for the no. of defects using 5 classes.
ii) Construct a boxplot for the no of defects in a part.
iii) ~~Constru~~ Compute the mean, median & the mode for the no of defects in a part.



$$\text{Mean} = \frac{2*30+3*15+4*9+5*4+6*2+7*1+10*1}{62}$$

$= 190/62 = 3.06$.

Mode = 2. (highest occuring).

Cum frequency 30 45 54 58 60 61 62.

$\frac{n+1}{4} = \frac{62+1}{4}$

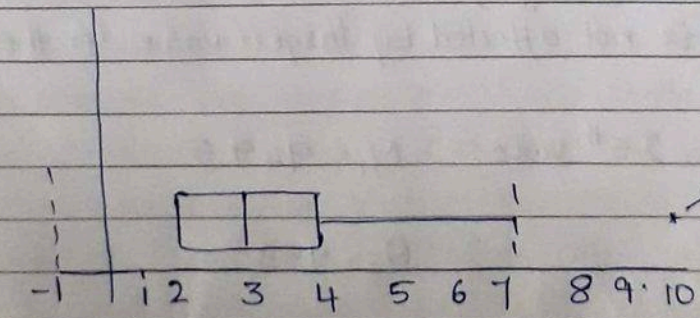
$= \frac{63}{4} = 15.75$

$Q_1 = 15.75$ $Q_2 = 31.5$ $Q_3 = 47.25^{th}$ position

$Q_1 = 2$ $Q_2 = 3$ $Q_3 = 4$. IQR = 4-2.

Fence = 2-1.5*2 = -1.

4+1.5*2 = 7.



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3. The nine measurements that follow are furnace temp recorded on successive batches in a semiconductor manufacturing process (units are in °F)

953, 950, 948, 955, 951, 949, 957, 954, 955

a) Calculate the sample mean, variance and std. deviation.

b) Find the median. How much could the largest temp. measurement increase without changing the median value?

c) Construct a box plot of the data.

a) $\bar{X} = \frac{953+950+948+955+951+949+957+954+955}{9}$

$\simeq 952.444$

$S^2 \simeq \frac{(953-952.44)^2 + \cdots + (955-952.44)^2}{9-1} \simeq 9.53$

$S = \sqrt{9.53} = 3.0867$

b) For median calculation.

Sort in ascending order

948, 949, 950, 951, 953, 954, 955, 955, 957

No. of measurement is odd, so median is middle value of sorted data set.

$M = Q_2 = 953$.

There is no limit to which highest temp. measurement can increase without changing the median value, ∵ the middle value is not affected by largest value in the data set.

$Q_1 = \frac{n+1}{4} = \frac{10}{4} = 2.5^{th}$ value. $Q_1 = 949.5$.

$Q_2 = 5^{th}$ value $Q_2 = 953$

$Q_3 = 7.5^{th}$ position $Q_3 = 955$

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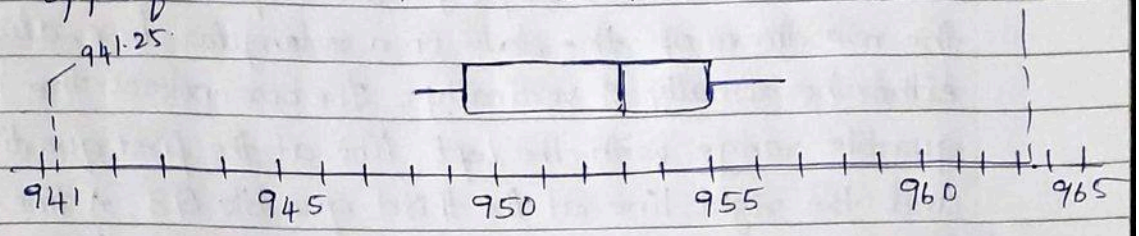
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Range = 957 - 948 = 9.

Inter quartile range = $Q_3 - Q_1$ = 955 - 949.5 = 5.5.

Lower fence = 949.5 - 1.5 * 5.5 = 941.25.

Upper fence = 955 + 1.5 * 5.5 = 963.25



[Histogram: classes should be all-inclusive & mutually exclusive.

* cumulative frequency & cum. relative frequency histogram called as Ogive curve.

classes ≥ 6-15, ≥ $\sqrt{n}$ for $n \leq 100$.

≥ $1 + 3.322 \log n$ for $n > 100$] → Sturges formula]

Stem and Leaf plot — Suppose that the data are represented by $x_1, x_2, \ldots x_n$ and that each number x_i consists of at least two digits. To construct a stem and leaf plot, we divide each number x_i into two parts: a stem, consisting of one or more of the leading digits, and a leaf consisting of remaining digits. Eg. if no. is 76, then we could divide into stem 7 & leaf 6. In gen, chose few stems as compared to no of observations. It is best to choose b/w 5 to 20 stems. Once a set of stems has been choosen, they are listed along Left hand margin of the display and beside each stem all leaves corresponding to the observed data values are listed in the order in which they are encountered in the data set.

In order to assist finding percentiles, we order the leaves by magnitude producing an "ordered stem and leaf plot"

If there are 100 nos, the median or middle observation is the average of 2 observns which with ranks 50 & 51 or.

The tenth percentile is observn with rank (10+11)/2 = 10.5

The first quartile is (25+26)/2 = 25.5th rank no

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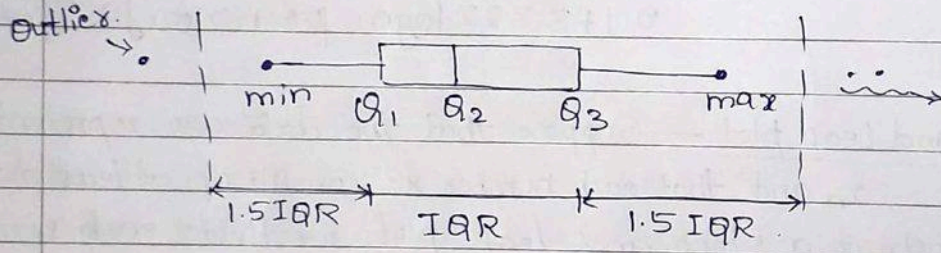
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Third quartile is 75.5^{th} rank no. The first & third quartile are occasionally denoted by the symbols Q_1 & Q_3 respectively and Interquartile range = $IQR = Q_3 - Q_1$

The Box Plot – It displays the 3 quartiles, the minimum, the maximum of the data on a rectangular box, aligned either horizontally or vertically. The box encloses the inter quartile range with the left line at the first quartile Q_1 and the right line at the third quartile Q_3. A line is drawn through the box at the second quartile (50 percentile from either end or median) $Q_2 = \bar{x}$. A line called whiskers extends to the extreme values. In large data sets it may extend only upto 10th & 90th percentile or 5th and 95th percentile.

Box plot is useful in comparing two or more samples.



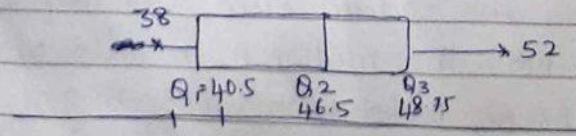
4. 38, 39, 41, 45, 46, 47, 48, 48, 51, 52

$n=10$ $Q_1 = \left(\frac{n+1}{4}\right)^{th}$ position, rank, value $= \left(\frac{11}{4}\right)^{th} = 2.75^{th}$ value $= 40.5$.

$Q_2 = \left(\frac{n+1}{2}\right)^{th}$ position $= 5.5^{th}$ value $= 46.5$

$Q_3 = \frac{3}{4}(n+1)^{th}$ value $= \frac{11}{4} \times 3 = 8.25^{th}$ value $= 48.75$.

$IQR = Q_3 - Q_1 = 48.75 - 40.5 = 8.25$.



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5. The number of defects when 20 parts were analysed were recorded as: 35, 18, 20, 29, 21, 8, 16, 14, 13, 25, 49, 21, 38, 18, 92, 19, 19, 15, 22, 29.

a) Draw the box plot.

b) Draw the stem and leaf diagram.

What inferences can you draw from these plots?

Stem	Leaf
0	8
1	86438995
2	0915129
3	58
4	9
5	
6	
7	
8	
9	2

Frequency	
1	1
8	9
7	(07)
2	4
1	2
1	1

Ordered.

Stem	Leaf
0	8
1	3 4 5 6 8 8 9 9
2	0 1 1 2 5 9 9
3	5 8
4	9
5	
6	
7	
8	
9	2

Inference – Large gaps in data, Min 8, Max - 92.

Box plot → Ascending order.

8, 13, 14, 15, 16, 18, 18, 19, 19, 20, 21, 21, 22, 25, 29, 29, 35, 38, 49, 92.

$Q_2 = \left(\frac{n+1}{2}\right)^{th}$ position = 10.5^{th} position = 20.5.

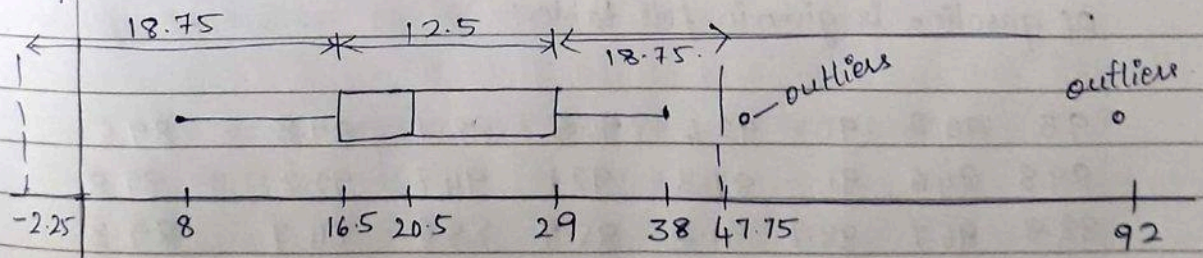
$Q_1 = \left(\frac{n+1}{4}\right)^{th}$ position = 5.25^{th} position = 16.5.

$Q_3 = \frac{3}{4}(n+1)^{th}$ = 15.75^{th} position = 29.

IQR = 29 - 16.5 = 12.50

Lower Fence = $16.5 - 12.5 \times 1.5 = -2.25$.

Upper Fence = $29 + 12.5 \times 1.5 = 47.75$.



Interpretation – Unsymmetric data set

Mid 50% data b/w 16.5 & 29. One suspected outlier 49 & one outlier - 92.

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6. 20 weight measurements in a process were recorded as
2, 8, 11, 10, 8, 4, 2, 2, 9, 8, 4, 11, 12, 2, 4, 4, 20, 1, 10, 15

(i) Draw a box plot for the data. Show the fences, quartiles, max and min clearly labelled in the sketch.
(ii) What are the range & the IQR for the data?
(iii) Are there any outliers? If yes, what are they?

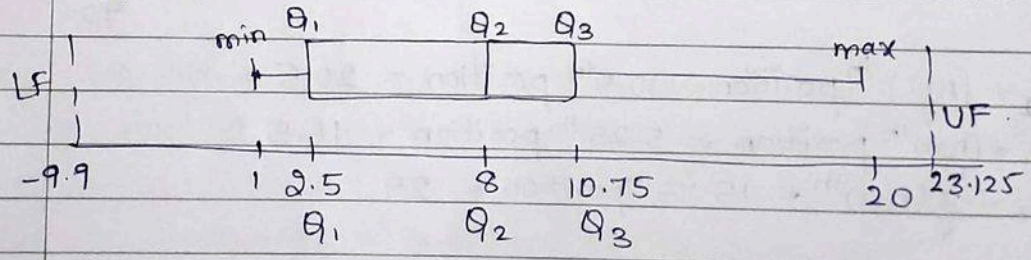
Sorted data:-

1 2 2 2 2 4 4 4 4 8 8 8 9 10 10 11 11 12 15 20.

$Q_1 = \frac{21}{4} = 5.25^{th}$ position Data 2.5
$Q_2 = (\frac{21}{2})^{th} = 10.5^{th}$ " $Q_2 = 8$.
$Q_3 = (\frac{3}{4} * 21)^{th} = 15.75^{th}$ " $Q_3 = 10.75$

Range = 20 − 1 = 19
IQR = 10.75 − 2.5 = 8.25

Lower fence = 2.5 − 1.5 * 8.25 = −9.875
Upper fence = 10.75 + 1.5 * 8.25 = 23.125



No outliers.

7. The data on the motor fuel octane ratings of several blends of gasoline is given in foll. table:

93	90.2	91	92.4	90.5	88.7	89.8	89.5
89.8	84.6	91	87.3	92.1	94.8	90.2	89.8
88.8	96.7	88.7	91.3	85.9	85.6	84.7	89.3
89.0	89.7	92.6	87.7	92.3	92.3	86.5	
89.5	87.3	88.7	94.1	87.2	85.4	90.7	

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Construct a stem and leaf diagram and highlight the important feature that can be noticed from the same.

Min- 84 Max 96

Stem	Leaf	Frequency
84	67	2
85	964	3
86	5	1
87	3732	4
88	7877	4
89	85883075	8
90	2527	4
91	003	3
92	41633	5
93	0	1
94	81	2
95		
96	7	1

ordered

Stem	Leaf	
84	67	2
85	469	5
86	5	6
87	2337	10
88	7778	14
89	03557888	(8)
90	2257	16
91	003	12
92	13346	9
93	0	4
94	18	3
95		
96	7	1

8. The nine measurements that follow are furnace temp recorded on successive batches in a semiconductor manfr. process (units are in °F)

953, 950, 948, 955, 951, 949, 957, 954, 955

a) Calculate the sample mean, sample variance and std. deviation.

b) Find the median. How much could the largest temperature measurement increase without changing the median value?.

c) Construct a box plot of the data.

$\bar{x} = 953+950+948+955+951+949+957+954+955 \,/\, 9$

$\simeq 952.444$

$s^2 = \dfrac{(953-952.44)^2 + \cdots + (955-952.44)^2}{9-1} \simeq 9.53.$

$s = \sqrt{9.53} = 3.0869.$

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Sorting in Ascending order.

948 949 950 951 953 954 955 955 957

No of records is odd, median is middle value of sorted data

M. $Q_2 = 953$

There is no limit to which highest temp. measurement can increase without changing the median value, bcoz the middle value is not affected by the largest value in the data set.

$Q_1 = 10/4 = 2.5^{th}$ position = 949.5.

$Q_2 = 953$.

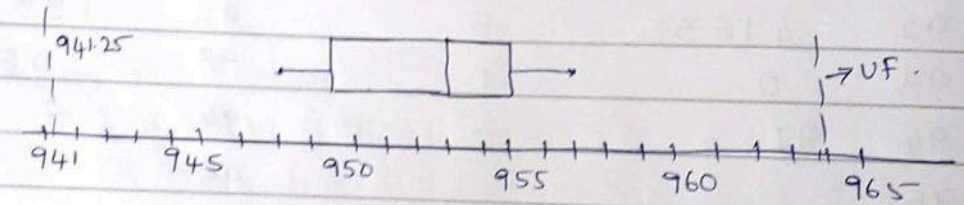
$Q_3 = 3/4 * 9 = 7.5^{th}$ position = 955.

IQR = 5.5.

Range = 957 - 948 = 9.

Lower fence = 949.5 - 1.5 * 5.5 = 941.25.

Upper fence = 955 + 1.5 * 5.5 = 963.25



9 Observe the stem & leaf diagram of history test scores and answer the foll.

July 2017

History test scores

Stem	Leaf
6	1 1 4 6 7 8
7	2 3 5 7 9
8	1 3 5 6 6 7 7 8 9
9	0 0 3 4 6 8 9 9
10	0 0

i) What is the best score? 100.

ii) How many students took the test? 30.

iii) How many students scored 90? 2.

iv) What is the lowest score? 61

v) Find the range for the scores = 100 - 61 = 39.

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10. Process times in a chemical industry are recorded in hours as follows:

4.2, 4.7, 4.7, 5.0, 3.8, 3.6, 3.0, 5.1, 3.1, 3.8, 4.8, 4.0, 5.2, 4.3, 2.8, 2.0, 2.8, 3.3, 4.8 & 5.0

i) Calculate the sample mean and std. dev.

ii) Construct a box plot of the data.

iii) Construct a stem & leaf plot & comment on the data

$\bar{x} = \frac{80}{20} = 4.0 \quad s^2 = \frac{16.46}{19} = 0.866 \quad s = 0.9307$

Stem	Leaf	freq.	Stem	Leaf (ordered)	
2	8 0 8	3	2	0 8 8	3
3	8 6 0 1 8 3	6	3	0 1 3 6 8 8	9
4	2 7 7 8 0 3 8	7	4	0 2 3 7 7 8 8	(7)
5	0 1 2 0	4	5	0 0 1 2	4

Stem → Tenths digit
Leaf → Ones digit

$Q_1 = \left(\frac{1}{4}\right)^{th} * 21 = 5.25^{th}$ position $= 3.15$

$Q_2 = 10.5^{th}$ position $= 4.1$

$Q_3 = 15.75^{th}$ position $= 4.8$ $\quad IQR = 4.8 - 3.15 = 1.65$

$L.F = 3.15 - 1.5 * 1.65$

$U.F = 4.8 + 1.5 * 1.65$

11. The foll. data represents variations in body parameters of 50 Asian male samples. Construct stem & leaf plot and compute the measures of central tendency.

28	56	68	58	34	74	52	37	62	98
33	40	48	08	34	65	37	19	46	76
16	70	99	35	67	61	07	10	29	94
72	66	26	28	85	17	14	39	66	03
16	79	23	92	09	30	73	55	23	46

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Stem	Leaf
0	8 7 3 9
1	9 6 0 7 4 6
2	8 9 6 8 3 3
3	4 7 3 4 7 5 5 9 0
4	0 8 6 6
5	6 8 2 5 5
6	8 2 7 1 6 6
7	4 6 0 2 9 3
8	
9	8 9 4 2

Ordered stem & leaf

Stem	Leaf
0	3 7 8 9
1	0 4 6 6 7 9
2	3 3 6 8 8 9
3	0 3 4 4 5 5 7 7 9
4	0 6 6 8
5	2 5 5 6 8
6	1 2 6 6 7 8
7	0 2 3 4 6 9
8	
9	2 4 8 9

Measure of central tendency Mean = 45.46.
Median = 39.5.

Radar chart

A radar chart is a graphical method of displaying multivariate data in the form of a two dimensional chart of three or more quantitative variables represented on axes starting from the same point.

Radar charts (also known as spider charts, polar charts, web charts or star plots) are a way to visualize multivariate data. They are used to plot one or more groups of values over multiple common variables.

The radar chart is a chart and/or plot that consists of a sequence of equiangular spokes, called radii, with each spoke representing one of the variables. The data length of a spoke is proportional to the magnitude of the variable for the data point relative to the max magnitude of the variable across all data points. A line is drawn connecting the data values for each spoke. This gives the plot a star-like appearance and the origin of one of the popular names.

This plot can be used to answer the foll. questions.

* Which observations are most similar ie are there clusters

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of observations? (Radar charts are used to examine the relative values for a single data point and to locate similar points or dissimilar points.

* Are there outliers?

Developing Radar charts

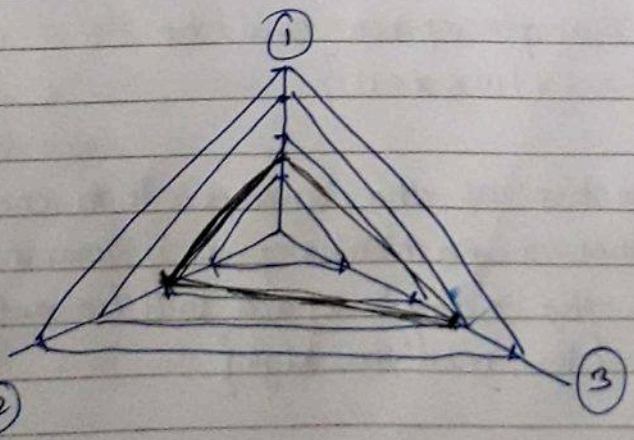
1. Identify the alternatives to be compared.
2. Generate criteria to rate each alternative.
3. Rate each alternative based on criteria.
4. Select a suitable structure & set relevant options. Set and label the axis.
5. Label the alternative's ratings on the chart.

Guidelines

1. No more than 15 alternatives to be chosen or else it will look cramped.
2. Use atleast 3 criteria, but more than 7 will become complex.
3. Standardize ratings to uniform scale.

Eg. Manufacturing process ratings.

Category	Our score	Best in class score	Standardize score
1. Safety	Reduced-5 accidents	Reduced-12 accidents	5/12 = 42%
2. Training expense.	Rs 10,000	Rs 20000	10,000/20,000 = 50%
3. Standard operating procedures.	74%	96%	74/96 = 77%



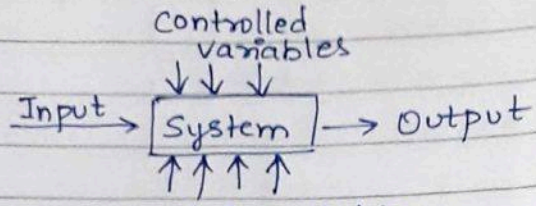
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Concepts of probability and Random variables



Random experiment → An experiment that can result in different outcomes, even though it is repeated in the same manner every time is called a random experiment.

Sample Spaces – The set of all possible outcomes of a random experiment is called the sample space of the experiment. The sample space is denoted by S
A sample space is often defined based on the objectives of the analysis.

Consides an expt in which you select a molded plastic part, such as a connector, and measure its thickness. The possible values for thickness depend on the resolution of the measuring instrument, & also depend on upper and lower bounds for thickness. It might be convenient to define the sample space as simply the positive real line ∵ a -ve value for thickness cannot occur.

$$S = R^+ = \{x \mid x > 0\}.$$

If it is known that all connectors will be between 10 & 11 mm thick, sample space could be

$$S = \{x \mid 10 < x < 11\}$$

If the objective of the analysis is to consider only whether a particular part is low, medium or high for thickness, the sample space will be set of 3 outcomes:

$$S = \{\text{low, medium, high}\}$$

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If the objective of the analysis is to consider only whether or not a particular part conforms to manufactoring specfns, the sample space

$S = \{yes, no\}$ that indicate whether or not the part conforms.

Discrete or Continuous Sample spaces -

A sample space is discrete if it consists of a finite or countable infinite set of outcomes.

A sample space is continuous if it contains an interval (either finite or infinite) of real numbers.

In previous examples the choice $S = R^+$ is an example of continuous sample space, whereas $S = \{yes, no\}$ is a discrete sample space.

Eg.	Expt.	Sample space
(1)	throw a die	$\{1, 2, \ldots 6\}$
(2)	Record claim amounts in all insurance policies	$\{0, 1, \ldots \infty\}$
(3)	Record ht. of students in cm	$\{x \mid 100 \leq x \leq 250\}$
(4)	Record of all accidents/day in front of RVCE gate	$\{0, 1, 2 \ldots 100\}$

The sample spaces that contain the outcomes of dice rolls, drawings from a bag of mixed color balls and dealings from a regular 52 card deck are examples of discrete sample spaces. Another eg is the no. of roulette wheel spins made before the ball lands on 25; the no. can range from 1, 2, 3 ... ∞, but no has to be integer.

Sample spaces that contain the outcomes of temp readings, ht. measurements and salaries are examples of continuous sample spaces.

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Eg.

Expt.	Sample space
(1) throw a die	$\{1, 2, \ldots 6\}$
(2) Record claim amounts in all insurance policies	$\{0, 1, \ldots \infty\}$
(3) Record ht. of students (m)	$\{x \mid 100 \leq x \leq 250\}$
(4) Record of all accidents/day in front of RVCE gate	$\{0, 1, 2 \ldots 100\}$

The sample spaces that contain the outcomes of dice rolls, drawings from a bag of mixed color balls and dealings from a regular 52 card deck are examples of discrete sample spaces. Another eg is the no. of roulette wheel spins made before the ball lands on 25; the no. can range from 1, 2, 3 ... ∞, but no has to be integer.

Sample spaces that contain the outcomes of temp readings, ht. measurements and salaries are examples of continuous sample spaces.

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In random experiments in which items are selected from a batch, we will indicate whether or not selected item is replaced before the next one is selected.
Eg. if the batch consists of 3 items $\{a, b, c\}$ & our expt is to select 2 items without replacement, the sample space can be represented as

$$S_{without} = \{ab, ac, ba, bc, ca, cb\}$$
$$\text{or } \{\{a,b\}, \{b,c\}, \{c,a\}\}$$

If items are replaced before the next one is selected, the sampling is referred to as with replacement. Then the possible ordered outcomes are

$$S_{with} = \{aa, ab, ac, ba, bb, bc, ca, cb, cc\}$$

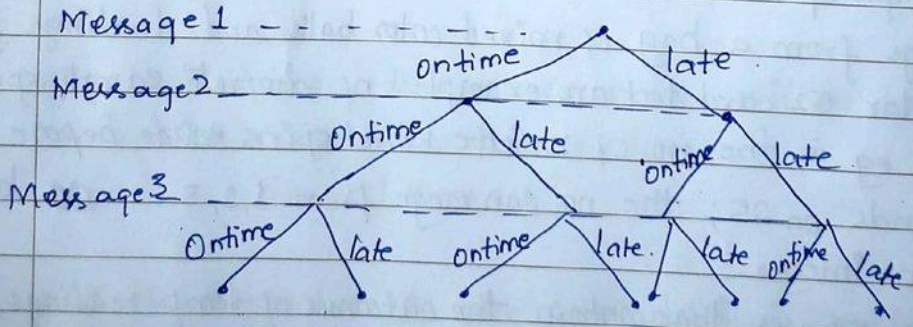
Other example, suppose that there are 5 defective parts & 95 good parts in a batch. To study the quality of the batch, 2 are selected without replacement. Let g denote a good part & d denote a defective part. Sample space (ordered) in this case

$$S = \{gg, gd, dg, dd\}$$

If one defective part in the batch, then sample space is

$$S = \{gg, gd, dg\}.$$

Sample spaces can also be described with tree diagrams.



Each msg in digital comm^n system is classified as to whether it is received within the time specified by system design. If 3 msgs are classified, the possible results of 3 msgs can be displayed by 8 branches in the tree diagram.

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An event is a subset of the sample space of a random expt.

UNION → of 2 events, is the event that consists of all outcomes that are contained in either of 2 events. $E_1 \cup E_2$

INTERSECTION of 2 events is the event that consists of all outcomes that are contained in both of the two events. $E_1 \cap E_2$.

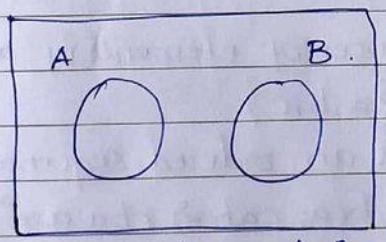
Complement of the event in the sample space is the set of outcomes in the sample space that are not in the event. Denote it as E', E^c

1. Measurements of the thickness of a plastic connector might be modeled with sample space $S = R^+$, the set of +ve real nos.
Let $E_1 = \{x \mid 10 \leq x < 12\}$ and $E_2 \{x \mid 11 < x < 15\}$

Then $E_1 \cup E_2 = \{x \mid 10 \leq x < 15\}$ and $E_1 \cap E_2 = \{x \mid 11 < x < 12\}$
Also $E_1' = \{x \mid x < 10 \text{ or } 12 \leq x\}$ and $E_1' \cap E_2 = \{x \mid 12 \leq x < 15\}$.

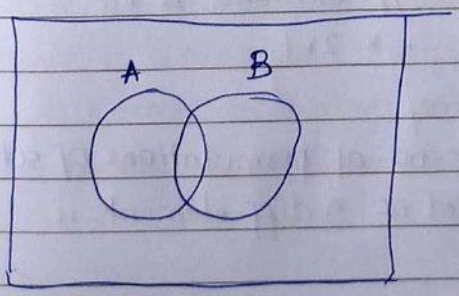
VENN DIAGRAMS are used to represent a sample space and events in the sample space.

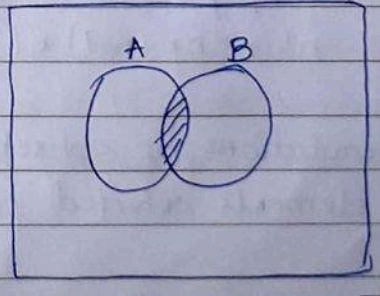
Two events, denoted as E_1 and E_2, such that
$E_1 \cap E_2 \neq \phi$ are said to be mutually exclusive.



Mutually exclusive events.

$A \cap B$





Sample space with events A & B

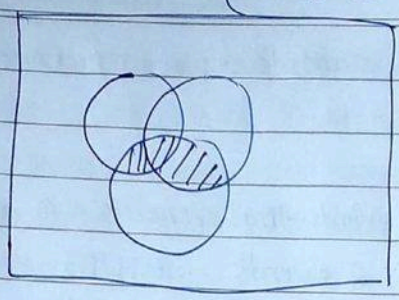
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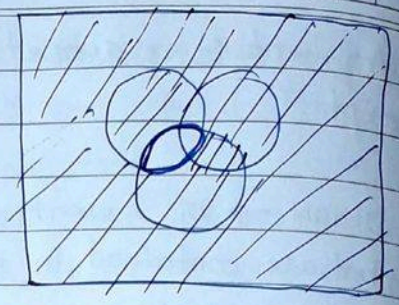
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$(A \cap C)'$

$(A \cup B) \cap C$ DATE





Complement of E complement is E $(E')' = E$

Distributive law for set opns

$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ and $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

DeMorgan's law imply that

$(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$

$A \cap B = B \cap A$ & $A \cup B = B \cup A$.

Counting Techniques → Det no of o/c in each event.

Multiplication Rule for counting technique.

If an opn can be described as sequence of k steps &

if no of ways of completing 1st step is n_1 &

" " " " " 2nd " is n_2 for each way of completing step 1 & so forth

→ the total no of ways of completing the operation is

$n_1 \times n_2 \times \cdots \times n_k$.

Permutation → ordered sequences of elements of a set.

Consider a set of elements $S = \{a, b, c\}$.

A Permutation of the elements is an ordered sequence of elements. For eg abc, acb, bac, bca, cab & cba are all permutation of element S.

The no of permutations of 'n' diff elements is n!

$n! = n * (n-1) * (n-2) * \cdots * 2 * 1$

Permutations of subsets — The no. of permutations of subsets of r elements selected from a set of n diff elements is

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$P_r^n = n*(n-1)*(n-2)*\cdots*(n-r+1) = \frac{n!}{(n-r)!}$

2. A printed circuit board has 8 diff locations in which a component can be placed. If four different components are to be placed on the board, how many diff designs are possible?

$P_4^8 = 8*7*6*5 = \frac{8!}{4!} = 1680$ different designs.

Permutations of similar objects — The no of permutations of $n = n_1 + n_2 + \cdots n_r$ objects of which n_1 are of one type, n_2 are of 2nd type & n_r are of rth type is

$$\frac{n!}{n_1! n_2! \cdots n_r!}$$

Problems. A part is labelled by printing with 4 thick lines, 3 medium lines and 2 thin lines. If each ordering of 9 lines rep a diff label, how many diff labels can be gen. by using this scheme?

No of possible part labels is

$$\frac{9!}{4!\,3!\,2!} = 1260.$$

Combinations — The no. of combinations, subsets of size r that can be selected from a set of n elements is denoted as $\binom{n}{r}$ or C_r^n and

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

4. A bin of 50 parts manufactured contain 3 defective parts and 47 non defective parts. A sample of six parts is selected from 50 parts and not replaced. How many different samples are these of size six that contains exactly 2 defective parts?

Soln. A subset containing exactly 2 defective parts can be formed by chosing 2 defective parts from 3 defective parts

$$3C_2 = \frac{3!}{1!\,2!} = 3 \text{ ways.}$$

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2nd step is to select the remaining 4 parts from 47 acceptable parts in bin.

$$^{47}C_4 = \frac{47!}{4!\,43!} = 178365 \text{ diff. ways}$$

By multiplication rule, the no. of subsets of size six that contain exactly 2 defective items is

$$3 * 178365 = 535,095$$

The total no of diff subsets of six 6 is

$$^{50}C_6 = \frac{50!}{44!\,6!} = 15,890,700.$$

$$\therefore \text{probability} = \frac{535,095}{15,890,700} = 0.034$$

$$= \frac{\text{no. of o/c in event}}{\text{" " " in samplespace}}.$$

Sample space examples

2-1 1. Each of 3 mlched parts is either classified as above or below target spefn for the part.

Let a, b denote a part above, below specfn respectively

$S = \{aaa, aab, aba, abb, baa, bab, bba, bbb\}$

2-3 2. In the final inspn of electronic power supplies, 3 types of nonconformities might occur, functional, minor or cosmetic. Power supplies that are defective are further classified as to type of nonconformity

a → acceptable power supply

f, m, c → defective category

$S = \{a, f, m, c\}$

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2-6 3. An ammeter that displays 3 digits is used to measure current in milliamperes.

The sample space is 1000 possible 3 digit integers

$S = \{000, 001, \ldots 999\}$

2-11 A. The pH reading of a water sample to the nearest tenth of a unit

$S = \{1.0, 1.1, 1.2, 1.3 \ldots 14.0\}$.

2-17 5. Calls are repeatedly placed to a busy phoneline until a connect is achieved. c = connect b = busy

$S = \{c, bc, bbc, bbbc, bbbbc, bbbbbc \ldots\}$.

2-22 6. In a injection molding opn, the length and width, denoted by X & Y respectively, of each molded part are evaluated. Let

A denote the event of $48 < X < 52$ cm.

B denote the even of $9 < Y < 11$ cm.

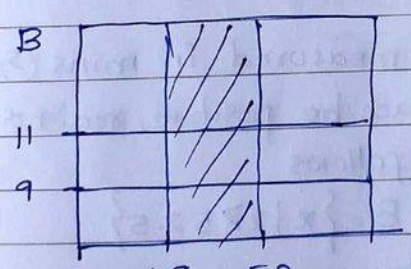
C. denote the event that a critical length meets customer requirements.

Construct a Venn diagram that includes these events. Shade the areas that represent the foll:

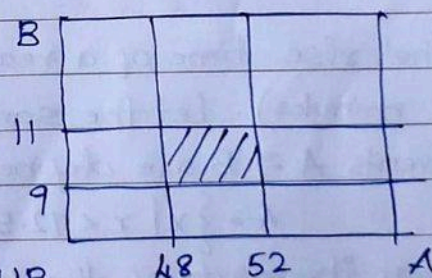
a) A b) $A \cap B$ c) $A' \cup B$ d) $A \cup B$

e) If these events were mutually exclusive, how successful would this prodn operation be? Would the process produce parts with X = 50 cms & Y = 10 cms?

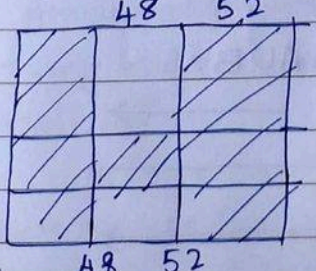
Soln.



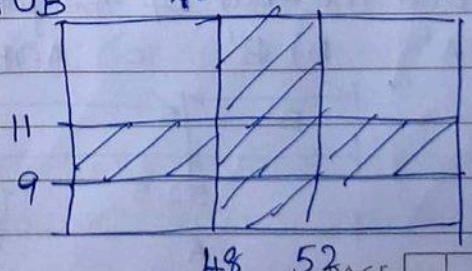
(A)



$A \cap B$



$(A' \cup B)$



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(e) If the events are mutually exclusive, then $A \cap B$ is null set
∴ the process does not produce parts with $X = 50cm$ & $Y = 10cm$. The process would not be successful.

7. Samples of emissions from 3 suppliers are classified for conformance to air-quality specifications. The results from 100 samples are summarized as follows:

Supplier	conforms: Yes	NO.
1	22	8
2	25	5
3	30	10

Let A denote the event that a sample is from supplier 1, & let B denote the " " " " conforms to specifications. Det. the no of samples in $A' \cap B$, B', and $A \cup B$.

Soln

$A' \cap B$ = sample not from supplier (1) & conforms to specifications

$= 25 + 30 = 55$

B' = sample not conforming $= 8 + 5 + 10 = 23$.

$A \cup B$ = Samples from (1) or & conforming
= samples from (1) + samples conforming − (samples from (1) & conforming)
$= (22 + 8) + (22 + 25 + 30) - 22$.

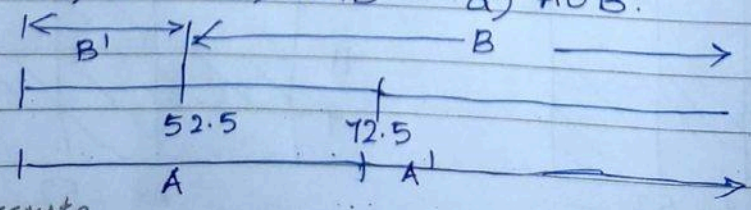
$A \cup B = 85$

2-29 8. The rise time of a reactor is measured in mins (& fraction of minutes). Let the sample space be positive, real nos. Events A & B are defined as follows

$A = \{x \mid x < 72.5\}$ & $B = \{x \mid x \geq 52.5\}$.

Describe each of the foll events.

(a) A' b) B' c) $A \cap B$ d) $A \cup B$.



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$A \cup B = S = \{ x \mid 0 < x < \infty \}$.

$A \cap B = S' = \{ x \mid 52.5 < x < 72.5 \}$.

2-31

9. A sample of 2 printed circuit boards is selected without replacement from a batch. Describe the (ordered) sample space for each of the foll batches:

a) The batch contains 90 boards that are not defective, 8 boards with minor defects & 2 boards with major defects

b) The batch contains 90 boards that are not defective, 8 boards with minor defects & 1 board with major defect.

Soln Let $\bar{d}$ = no defects.

d = minor defects.

D = major defects.

$S = \{ \bar{d}\bar{d}, \bar{d}d, \bar{d}D, d\bar{d}, dd, dD, D\bar{d}, Dd, DD \}$.

$S = \{ \bar{d}\bar{d}, \bar{d}D, \bar{d}d, d\bar{d}, dd, dD, D\bar{d}, Dd \}$.

2-35

10. An order for a personal digital assistant can specify any one of five memory sizes, any one of 3 types of displays, any one of 4 sizes of a hard disk, & can either include or not include a pen tablet. How many diff. systems can be ordered?

with r.t. memory sizes → 5 ways.

display → 3 ways.

Hard disks → 4 sizes.

Tablet → 2. (include/not include)

Multiplication rule → Total no of diff systems

$= 5 * 3 * 4 * 2 = 120$.

2-38

11. A manufacturing process consists of 10 opns that can be completed in any order. How many diff. production sequences are possible?

$10! = 10 * 9 * 8 * 7 * 6 * 5 * 4 * 3 * 2 * 1 = 3628800$.

Where we use the fact that when we select one opn, it cannot be selected again.

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2-41

12. A lot of 140 semiconductor chips is inspected by chosing a sample of five chips. Assume 10 of the chips do not conform to customer requirements.

a) How many different samples are possible?

b) How many samples of 5 contain exactly one nonconforming chip?

c) How many samples of 5 contain atleast one nonconforming chip?

Soln

a) No of diff samples are

$$140C_5 = \frac{140!}{(135)!(5)!} = 41,69,65,528.$$

b) No of samples of 5 containing exactly 1 nonconforming chip

$$= (130C_4)(10C_1) = \frac{130!}{126!\,4!} * \frac{10!}{9!*1!}$$

$$= 11,35,88,800.$$

c). No of samples of 5 containing atleast one nonconforming chip

= (4 good & 1 bad) or (3 good and 2 bad) or (2 good & 3 bad) or (1 good & 4 bad) or (0 good & 5 bad)

$$= \binom{130}{4}\binom{10}{1} + \binom{130}{3}\binom{10}{2} + \binom{130}{2}\binom{10}{3} + \binom{130}{1}\binom{10}{4} + \binom{130}{0}\binom{10}{5}$$

$$= 113588800 + 16099200 + 10,06200 + 27300 + 252$$

$$= 130,721,752$$

2-44

13. In the design of an electromechanical product, 7 diff components are to be stacked into a cylindrical casing that holds 12 components in a manner that minimizes the impact of shocks. One end of the casing is designated as the bottom & the other end is the top.

(a) How many diff. designs are possible?

(b) If the seven components are all identical, how many diff. designs are possible?

c) If the 7 compo. consists of 3 of one type of component & 4 of another type, how many different designs are possible

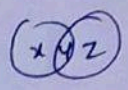
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Addition rule:

$P(A \cup B) = x + y + z$

$= (x+y) + (y+z) - y$

$= P(A) + P(B) - P(A \cap B)$



DATE

(a) $12! = 479,001,600 = (12C_7) * 7! = \frac{12!}{7! * 5!} * 7!$

$= 59,91,680.$

(b) Since $12 - 7 = 5$ components are diff, we have $n_1, \ldots n_5$ are equal to 1 & 7 are identical, therefore $n_6 = 7$

$$\frac{12!}{1! * 1! * 1! * 1! * 1! * 7!} = 95,040$$

c). 3 are one type, 4 are of another type then.

$$\frac{12!}{1! * 1! * 1! * 1! * 1! * 3! * 4!} = 33,26,400 \times$$

$$= (12C_7) * \frac{7!}{3!\,4!} = \frac{12!}{5! * 3! * 4!} = 27,720$$

Equally likely outcome – Whenever a sample space consists of N possible outcomes that are equally likely, the probability of each outcome is $1/N$.

For a discrete sample space, the probability of an event E, denoted as $P(E)$, equals the sum of the probabilities of the outcomes in E.

Axioms of Probability

Probability is a no that is assigned to each member of a collection of events from a random experiment that satisfies the foll. properties:

If S is the sample and E is any event in a random experiment,

(1) $P(S) = 1$

(2) $0 \leq P(E) \leq 1$.

(3) For two events E_1 & E_2 with $E_1 \cap E_2 = \emptyset$

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

$$P(\emptyset) = 0$$

and for any event E

$$P(E') = 1 - P(E)$$

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14 If the last digit of a wt. measurement is equally likely to be any of the digits 0 through 9,

a) What is the probability that the last digit is 0?

b) What is the probability that the last digit is > or = 5?

a) $1/10 = 0.1$

b) $5/10 = 0.5$.

15. A credit card contains 16 digits between 0 and 9. However only 100 million numbers are valid. If a no. is entered randomly, what is the probability that it is a valid number.

16 digits b/w 0 and 9.

We have – total no of possibilities

$$N_T = (\text{No. of states for digit})^{\text{No of digit}} = 10^{16}.$$

Let A rep the event that the random no entered is a valid no

$$n(A) = 100 * 10^6.$$

$$\text{probability} = \frac{n(A)}{N} = \frac{100*10^6}{10^{16}} = \frac{1}{10^8} = 1*10^{-8}$$

16. Suppose your vehicle is licensed in a state that issues license plates that consist of 3 digits (b/w 0 and 9) followed by 3 letters (between A and Z). If a licence no is selected randomly, what is the prob that yours is the one selected.

10 digits 3 letters $N_D = 10^3$.

$N_A = 26^3$

$$\text{prob} = \frac{1}{10^3 * 26^3}$$

(17) Problem a) P(A) b) P(B) c) P(A') d) P(A ∩ B) e) P(A∪B) f) P(A'∪B)

a) P(A) = 30/100 = 0.3

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b) $P(B) = 77/100 = 0.77$
c) $P(A') = 0.7$
d) $P(A \cap B) = 22/100 = 0.22$
e) $P(A \cup B) = 30 + 77 - 22 = 85/100 = 0.85$.
f) $P(A' \cup B) = 70 + 77 - 55 = 92/100 = 0.92$.

18. Disks of polycarbonate plastic from a supplier are analysed for scratch & shock resistance. The results from 100 disks are summarized as follows.

		Shock resistance	
		high	low
Scratch resistance	high	70	9
	low	16	5

Let A denote the event that a disk has high shock resistance, & let B denote the event that a disk has " scratch ". If a disk is selected at random, det the foll prob.
a) $P(A)$ b) $P(B)$ c) $P(A')$ d) $P(A \cap B)$ e) $P(A \cup B)$
f) $P(A' \cup B)$

(a) $P(A) = 86/100 = 0.86$ b) $P(B) = 0.79$.
(c) $P(A') = 14/100 = 0.14$ d) $P(A \cap B) = 0.70$.
(e) $P(A \cup B) = 0.86 + 0.79 - 0.70 = 0.95$.
(f) $P(A' \cup B) = 0.14 + 0.79 - 0.09 = 0.84$.

ADDITION RULES.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A & B are mutually exclusive
$P(A \cup B) = P(A) + P(B)$.

$$P(A \cup B \cup C) = P[(A \cup B) \cup C] = P(A \cup B) + P(C) - P[(A \cup B) \cap C]$$

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) + P(B \cap C) + P(A \cap B \cap C)$$

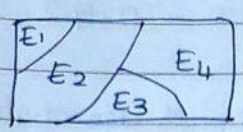
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A colln of events $E_1, E_2 \dots E_k$ is said to be mutually exclusive if for all pairs

$$E_i \cap E_j = \emptyset$$

Then $P(E_1 \cup E_2 \cup \dots \cup E_k) = P(E_1) + P(E_2) + \dots + P(E_k)$



4 mutually exclusive events.

19. Customers order any combinations of items A, B, C

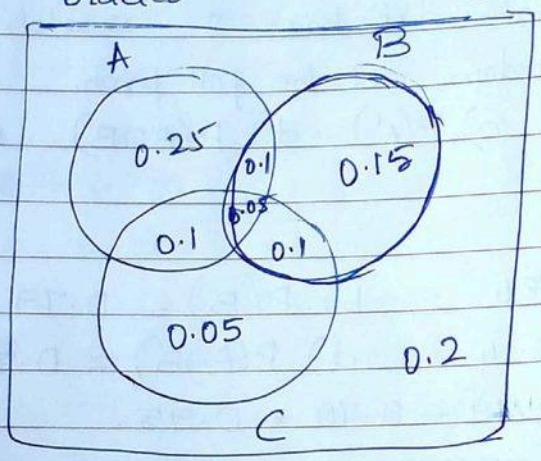
$P(C)_{\text{ordered}} = 0.3$

$P(B) = 0.4$

$P(A \& B)_{\text{ordered}} = 0.15$ $\quad P(A \& C)_{\text{ordered}} = 0.15$

$P(B \& C)_{\text{ordered}} = 0.55$

$P(A, B \& C)_{\text{ordered}} = 0.05$ $\quad P(\text{None})_{\text{ordered}} = 0.2$



$P(A) = ?$

$P(A \text{ alone})_{\text{ordered}} = ?$

$$P(B \text{ or } C) = P(B \cup C) = P(B) + P(C) - P(B \cap C)$$

$$0.55 = 0.4 + 0.3 - P(B \cap C)$$

$$\Rightarrow P(B \cap C) = 0.7 - 0.55 = 0.15 = P(B \& C \text{ ordered})$$

$$P(A \text{ alone ordered}) = 1 - 0.2 - (0.3 + 0.15 + 0.1) +$$

$$= 0.25.$$

$$\Rightarrow P(A) = 0.25 + 0.1 + 0.1 + 0.05 = 0.5$$

20. A manufacturer of front lights for automobile tests lamps under a high humidity, high temp envn. using intensity & useful life as the responses of interest. The foll table shows the performance of 130 lamps

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Intensity		Useful life: satisfactory	Useful life: unsatisfactory
	satisfactory	117	3
	unsatisfactory	8	2

(i) Find the prob that a randomly selected lamp will yield unsatisfactory results under any criteria.

ii) The customers for these lamps demand 95% satisfactory results. Can the lamp manufacturer meet this demand.

(i) = 3+2+8 = 13/130 = 0.1.

(ii) 90% satisfactory - so no the lamp manufacturer has not met this demand.

CONDITIONAL PROBABILITY.

The conditional probability of an event B given an event A, denoted as P(B/A) is

$$P(B/A) = P(A \cap B) / P(A) \quad \text{for } P(A) > 0$$

This definition can be understood in a special case in which all outcomes of a random experiment are equally likely. If there are n total outcomes

P(A) = (no. of o/c in A)/n.

P(A∩B) = (no of o/c in A∩B)/n.

$$\therefore P(A \cap B) / P(A) = \frac{\text{no. of outcomes in } A \cap B}{\text{no. of outcomes in } A}.$$

21. A day's production of 850 manufactured parts contains 50 parts that do not meet customer requirements. Two parts are selected randomly without replacement from the batch. What is the probability that the second part is defective given that the first part is defective?

Let A denote the event that the 1st part is defective

Let B " " " " " 2nd " " "

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$P(B/A) = 49/849$.

If 3 parts are selected randomly, what is prob that 1st two are defective and third is not defective?

$p(ddn) = \frac{50}{850} * \frac{49}{849} * \frac{800}{848} = 0.0032$.

22. 18th problem continued

$P(A|B) = 0.70/0.79 = 0.886$.

$P(B/A) = P(A \cap B)/P(A) = .70/0.86 = 0.814$.

2-82 23 A maintenance firm has gathered the foll information re the failure mechanisms for air conditioning systems.

		evidence of gas leaks: yes	no
evidence of electrical failure	yes	55	17
	no	32	3

The units without evidence of gas leaks or electrical failure other types of failure. If this is a rep. sample of AC failure the prob

a) That failure involves a gas leak.
b) That there is evidence of electrical failure given that there was a
c) " " " " " gas leak given that there is evi of electrical failure

Let E represent electrical failure
G - gas leak.

Given.

	G	G'	
E	55	17	$P(E) = 72/107 = 0.67$.
E'	32	3	$P(E') = 35/107 = 0.33$.

$P(G) = 87/107 = 0.81$ $P(G') = 20/107 = 0.19$

(i) $P(G) = 87/107 = 0.81$ (failure involves gas leak

(ii). $P(E/G) = P(E \cap G)/P(G) = \frac{55/107}{87/107} = \frac{55}{87}$

(iii) $P(G/E) = P(G \cap E)/P(E) = 55/72$

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2.4

2.76. Let A → low melatrin content.
B → high moisture content.

		Melanin content	
		high	low
Moisture content	high	13	7
	low	48	32

a) $P(A) = 39/100 = 0.39$
b) $P(B) = 20/100 = 0.2$
c) $P(A/B) = P(B \cap A)/P(B)$
$= 7/20 = 0.35$
d) $P(B/A) = P(A \cap B)/P(A)$
$= 7/39$

2.78

25. Samples of a cast aluminum part are classified on the basis of surface finish (in microinches) and length measurements. The results of 100 parts are summarized as follows:

		Length	
		excellent	good
Surface finish	excellent	80	2
	good	10	8

Let A denote the event that the sample has excellent surface finish, Let B denote the event " has excellent length.

a) P(A) b) P(B) c) P(A|B) d) P(B/A)

e) If selected part has excellent surface finish, what is prob. that the length is excellent.

f) If selected part has good length, what is the prob that S. finish is excellent.

Soln $P(A) = 82/100 = 0.82$ $P(B) = 90/100 = 0.90$.

$P(A/B) = P(A \cap B)/P(B) = 80/90 = 0.889$.

$P(B/A) = P(A \cap B)/P(A) = 80/82 = 0.976$.

$P(B/A) = 0.976$.

$P(A/B') = P(A \cap B')/P(B') = 2/10 = 0.2$.

2.81

26 The following table summarizes the no. of deceased beetles under autolysis (the destruction of a cell after its death by the action of its own enzymes) and putrefaction (decomposition of organic matter, especially protein by microorganisms, resulting in prodn of foul smelling matter):

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		autolysis high	autolysis low
putrefaction	high	14	59
	low	18	9

$P(P) = 73/100$ $P(P') = 27/100$

a) If the autolysis of a sample is high, what is the prob that putrefaction is low?

b) If the putrefaction of a sample is high, what is the prob autolysis is high?

c) If the putrefaction of a sample is low, what is the prob tha autolysis is low?

Let A → Autolysis high. $P(A) = 32/100$.

P → putrefaction high. $P(A') = 68/100$.

a) $P(P'/A) = P(P' \cap A)/P(A) = 18/32 = 0.5625$.

b) $P(A/P) = P(A \cap P)/P(P) = 14/73 = 0.1918$.

c) $P(A'/P') = P(A' \cap P')/P(P') = 9/27 = 0.333$

2-83

27 A lot of 100 semiconductor chips contains 20 that are defect Two are selected randomly, without replacement from the lot

a) What is the prob. that the 1st one selected is defective?

b) " " " " " " 2nd " " " " " " agai given that the first one is defective?

c) What is the probability that both are defective?

d) How does the answer to part (b) change if chips selected were replaced prior to the next selection?

Soln (a) $20/100 = 0.2$.

(b) $P(\text{II d}/\text{I d}) = \frac{\text{No of defective items}}{\text{Total}} = 19/99$.

(c) $P = \frac{20}{100} * \frac{19}{99} = 0.03838$.

(d) $P = 20/100 =$ (if defective chip is replaced)

2-86.

28

(a) Password → exactly seven characters & each characters is one of 26 letters (a-z) & 10 integers (0-9). Let A denote the subset of passwords that begin with a vowel & let B denote the subset of passwords that end with an even no (0, 2, 4, 6, 8)

CLASSMATE

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a. Suppose a hacker selects a password at random. What is the prob that your password is selected.

b. Suppose a hacker knows your password is in event A & Selects a password at random from this subset. What is prob. that ur password is selected?

c. Suppose a hacker knows your password is in A & B & selects a password at random from this subset. Find pr.

(a) $P(\text{mine}) = \frac{1}{36^7}$

(b) $N_a = 5 * 36^7$.

$\therefore$ probability $(\text{mine}/A) = 1/5*36^7$.

(c) $N_c = 5 * 5 * 36^7$

$P(\text{mine}/AB) = \frac{1}{N_c} = \frac{1}{25 * 36^7}$

Multiplication rule.

$P(A \cap B) = P(B/A)\,P(A) = P(A/B)\,P(B)$.

Total probability rule

For any events A & B.

$P(B) = P(B \cap A) + B(B \cap A') = P(B/A)\,P(A) + P(B/A')\,P(A')$

Total prob. Rule (multiple events)

Assume $E_1, E_2 \ldots E_k$ are k mutually exclusive & exhaustive sets.

$P(B) = P(B \cap E_1) + P(B \cap E_2) + \cdots + P(B \cap E_k)$

$= P(B/E_1)P(E_1) + P(B/E_2)\,P(E_2) + \cdots + P(B/E_k)(P(E_k))$

2-91

29 The probability is 1% that an electrical connector that is kept dry fails during the warranty period of a portable computer. If a connector is ever wet the prob. of a failure during warranty period is 5%. If 90% of the connectors are kept

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dry and 10% are wet, what proportion of connectors fail during the warranty period.

Soln Event B → connectors fail A → wet condition failure = 5%. A' → dry " " = 1%.

$P(B) = P(B \cap A) + P(B \cap A')$

$= \frac{P(B/A)}{P(A)} \times + \frac{P(B/A')}{P(A')} \times = \frac{0.05}{0.9} + \frac{0.01}{0.1}$

$= 0.055 + 0.1$

$= P(B/A) * P(A) + P(B/A') * P(A')$

$= 0.05 * 0.9 + 0.01 * 0.9 = 0.014$

2-93
30 The edge roughness of slit paper products increases as knife blades wear. Only 1% products slit with new blades have rough edges, 3% of products slit with blades of avg sharpness exhibit roughness & 5% of products slit with worn blades exhibit roughness. If 25% of the blades in manfr are new, 60% are of avg sharpness & 15% are worn, what is the proportion of products that exhibit edge roughness?

Event B → rep. edge roughness.

A_1 → new blades with rough edges. $P(A_1) = 0.01$

A_2 → avg sharpness exhibit " $P(A_2) = 0.03$

A_3 → worn blades " " $P(A_3) = 0.05$.

$P(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3)$

$= P(B/A_1) * P(A_1) + P(B/A_2) * P(A_2) + P(B/A_3) * P(A_3)$

$= 0.01 * 0.25 + 0.03 * 0.6 + 0.05 * 0.15$

$= 0.028$

2-94
31 B → respondent voted for Bush. If 2020 respondents

Prob of voting to bush $P(B) = 0.62 * 0.50 + 0.38 * 0.53 = 0.5114$

$= \frac{1}{1033} = 0.00096.$

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2.95

32 Computer keyboard failures are due to faulty electrical connects (12%) or mechanical defects (88%). Mechanical defects are related to loose keys (27%) or improper assembly (73%) Electrical connect defects are caused by defective wires (35%) Improper connections (13%) or poorly welded wires (52%)

a) Find the probability that a failure is due to loose keys.

b) " " " " " " " " improperly connected or poorly welded wires.

E → failures electrical connect.

M → " Mechanical defects.

L → loose keys. A → improper assembly.

E → D.W + I.M (I: connection) + (P: Poorly welded)

P(L) = 0.27 * 0.88 = 0.2376.

~~P(P) = 0.52 * 0.12~~

P(IP) = P(Improper connection + Poor weld) =

0.52 * 0.12 + 0.13 * 0.12 = 0.0078

2.96

33 Heart failures are due to either natural occurrences (87%) or outside factors (13%). Outside factors are related to induced substances (73%) or foreign objects (27%). Natural occurrences are caused by arterial blockage (56%), disease (27%) & infection (17%)

a) Det. the prob. that a failure is due to induced substance.

b) " " " " " " " " disease or infection.

a) P(I.S) = 0.73 * 0.13 = 0.0949

b) P(D or I) = 0.27 * 0.87 + 0.17 * 0.87 = 0.3828.

INDEPENDENCE

Two events are independent if any one of the foll. equivalent statements is true:

1) P(A|B) = P(A) 2) P(B|A) = P(B)

3) P(A ∩ B) = P(A) P(B)

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2-107
34 The probability that a lab specimen contains high level of contamination is 0.10. Five samples are checked and the samples are independent.

(a) What is the probability that none contains high levels of contamination?

(b) What is the probability that exactly one contains high level of contamination?

(c) What is the probability that at least one contains high levels of contamination.

A_i → be the event that specimen contains high level of contamination

$P(A) = 0.1$.

Let A_i = the ith lab specimen contains high " " "

$P(A_i) = 0.1$

(a) Prob Event can be rep as

$$P(A_1' \cap A_2' \cap A_3' \cap A_4' \cap A_5') = P(A_1') * P(A_2') * P(A_3') * P(A_4')$$
$$= 0.9 * 0.9 * 0.9 * 0.9 * 0.9$$
$$= 0.5905$$

(c) It is $1 - P(A_1' \cap A_2' \cap A_3' \cap A_4' \cap A_5')$

$= 1 - 0.5905 = 0.4095 = 0.41$.

(b) Event that exactly one contains high levels of contamination can be rep. as disjoint union of events

$$\left(\bigcap_{i \neq i_0} A_i'\right) \cap A_{i_0}$$

for $i_0 = 1, 2, 3, 4, 5$. So the intersection of 5 events from which one of them contains high levels of contamination. The prob of each of the 5 events can be obtained the same way.

For eg for $i_0 = 2$, the foll is

$$P\left[\bigcap_{i \neq 2} A_i'\right) \cap A_2 = P(A_1' \cap A_3' \cap A_4' \cap A_5' \cap A_2)$$
$$= P(A_1') * P(A_3') * P(A_4') * P(A_5') * P(A_2)$$
$$= 0.9 * 0.9 * 0.9 * 0.9 * 0.1 = 0.066.$$

Since we need to find the prob of union of five ~~distinct~~ disjoint events with prob 0.066 the prob

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$P(E_1) = P\{\text{exactly one contains high level of contamination}\}$;

$= P\left[\bigcup_{i_0=1}^{5}\left(\left(\bigcap_{i\neq i_0} A_i'\right)\cap A_{i_0}\right)\right]$

$= P\left[\left(\bigcap_{i\neq 1} A_i'\right)\cap A_1\right] + \cdots + P\left[\left(\bigcap_{i\neq 5} A_i'\right)\cap A_5\right]$

$= 0.066 + 0.066 + 0.066 + 0.066 + 0.066 = 0.33.$

35 In an acid base titration, a base or acid is gradually added to the other until they have completely neutralized each other. Since acids and bases are usually colorless, pH is measured to monitor the reaction. Suppose that the equivalence point is reached after approx. 100 mL of a NaOH soln. has been added (enough to react with all the acetic acid present) but that replicates are equally likely to indicate from 95 to 104 mL, measured to the nearest mL. Assume that two technicians each conduct titrations independently.

(a) What is the prob that both technicians obtain equivalence of at 100 mL?

(b) What is the prob that both technicians obtain equivalence between 98 & 104 mL (inclusive)?

(c) What is the prob. that the avg volume at equivalence from the technicians is 100mL?

Soln Technicians T_1 & T_2 independently obtain equivalence at foll qty

$Q = \{95, 96, 97, 98, 99, 100, 101, 102, 103, 104) = (q_i)_{i=1,\ldots,10}$

and with equal prob. Let t_1 & t_2 rep quantities at which technicians T_1 & T_2 obtain equivalence resp. This means

$P(t_1 = q_i) = \frac{1}{10}, \quad i = 1, \ldots, 10.$

$P(t_2 = q_i) = \frac{1}{10}, \quad i = 1, \ldots, 10.$

(a) $P(\{t_1 = 100\} \cap \{t_2 = 100\}) = P(\{t_1 = 100\})\, P(\{t_2 = 100\}) = \frac{1}{10} * \frac{1}{10} = \frac{1}{100}.$

(b) $P(\{t_1 \text{ and } t_2 \in \{98, \ldots 104\}\}) = P(t_1 \in \{98, \ldots, 104\}) * P(t_2 \in \{98, \ldots, 104\})$

$= \frac{7}{10} * \frac{7}{10} = \frac{49}{100}.$

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c. Total no of outcomes for obtaining equivalence is $10*10=1$
∵ both technicians can obtain equivalence in 10 possible wa
We need to find the no of relevant outcomes, i.e we need
all pairs (t_1, t_2) such that

$$\frac{t_1+t_2}{2} = 100$$

(100,100), (101,99), (99,101), (102,98), (98,102), (103,97)
(97,103), (104,96), (96,104)

∴ prob that avg volume at equivalence from technician
100mL

$$= 9/100 = 0.09$$

2109 36 Six tissues are extracted from an ivy plant infested by spider mites. The plant is infested in 20% of its area. A sample is chosen from randomly selected areas on the ivy plant. Assume that the samples are independent.

a) What is the prob. that four successive samples shows the signs of infestation?

b) What is the prob. that 3 out of 4 successive samples show the signs of infestation?

Soln A_i = (i^{th} tissue extracted from an ivy plant is infested by spider mi
$i=1,2\cdots6$. plant is infested in 20% of its area,

$$P(A) = 0.2 \quad \text{and} \quad P(A_i') = 1-0.2 = 0.8 \text{ for } i=1,2\cdots6$$

Let B = {four successive sample show the signs of infestation}

G - good
I - infestation IIIIGG, GIIIIG, GGIIII,

Prob. of union of disjoint events

$$P(A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5' \cap A_6') = P(A_1) * P(A_2) * P(A_3) \cdots P(A_6')$$

$$= 0.2 * 0.2 * 0.2 * 0.2 * 0.8 * 0.8$$

$$= 0.2^4 * 0.8^2 = 0.001024$$

$= 0.2^4 * 3 = 0.0048$

IIIIGG
GIIIIG
GGIIII

$$P(B) = 0.001024 + 0.001024 + 0.001024 = 0.003072$$

b. We can select 4 samples successively in 3 ways picking 3 out of them
can be done in $^4C_3 = 4$ ways. ∴ $n_1 * n_2 = 6$ diff. events
But 2 are counted. ∴ 10 diff events.

(1) IIIG, IIGI, IGII, GIII

$$P(A_1 \cap A_2 \cap A_3 \cap A_4' \cap A_5' \cap A_6') = 0.2^3 * 0.8^3 =$$

$$(0.2)^3 * (0.8)^1 * 12. = 0.0768$$

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2-110 Soln 37. When a player is defeated by opponent, then the game ends.

$P(\text{win}) = 80\% = 0.80 \quad n = 4$

$P(A \text{ and } B) = P(A) * P(B)$

Given is that the results from opponents are independent. The player defeats all 4 opponents if he wins each of 4 games.

(a) ∴ Probability that a player defeats all 4 opponents in a game $= 0.8 * 0.8 * 0.8 * 0.8$

$= 0.8^4 = 0.4096.$

(b) All possible game sequence (W = win & L = lose)

L, WL, WWL, WWWL, WWWW

We not that we win at least 2 times in last 3 sequences.

$P(A') = 1 - P(A)$

$P(L) = 1 - P(\text{win}) = 0.20.$

Probability of a win & then lose

$P(WL) = P(\text{win}) * P(\text{Lose}) = 0.8 * 0.2 = 0.16.$

Prob (At least 2 wins) $= 1 - 0.2 - 0.16 = 0.64.$

or $= \text{Prob}(WWL + WWWL + WWWW) = 0.128 + 0.1024 + 0.4096 = 0.64.$

(c) Prob. of not having 4 wins

$P(\text{not 4 wins}) = 1 - P(\text{4 wins}) = 1 - 0.4096 = 0.5904.$

$P(\text{3 times not 4 wins}) = P(\text{not 4 wins}) * P(\text{not 4 wins}) * P(\text{not 4 wins})$

$= 0.5904 * 0.5904 * 0.5904 = 0.2058.$

∴ $P(\text{at least once 4 wins}) = 1 - P(\text{3 times not 4 wins})$

$= 1 - 0.2058 = 0.7942.$

Problem 38

N = one million

2-112. (a) $= \frac{1 * 10^6}{10^{16}} = 10^{-10}.$

(b) Let B denote the event that hacker correctly selects month & yr of credit card expiration. $= P(B_1) = 0.25.$

A hacker randomly selects 1 of 12 months, so the prob of correctly guessing is 1/12.

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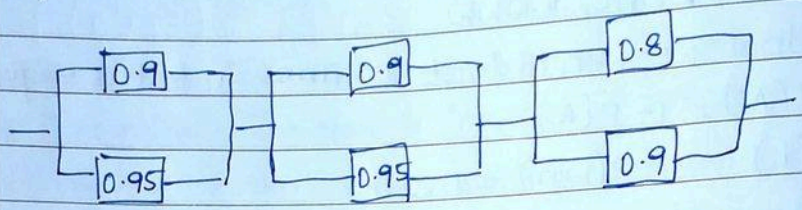
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The prob that hacker correctly selects the month & yr of credit card expiration is

$$P(B) = 0.25 * \frac{1}{12} = \frac{1}{4} * \frac{1}{12} = \frac{1}{48}.$$

39
2-115 The foll circuit operates if and only if there is a path of functional devices from left to right. The prob each device functions is as shown. Assume that the prob that a device fns. does not depend on whether or not other devices are functional. What is the prob. that the circuit operates?



$$= 0.995 * 0.995 * 0.98 = 0.9702.$$

BAYES THEOREM

If $E_1, E_2, \ldots E_k$ are k mutually exclusive and exhaustive events and B is any event,

$$P(E_1/B) = \frac{P(B/E_1)\,P(E_1)}{P(B/E_1)\,P(E_1) + P(B/E_2)\,P(E_2) + \cdots + P(B/E_k)P(E_k)}$$

for $P(B) > 0$.

40.
2-121 Customers are used to evaluate preliminary product designs. In the past, 95% of highly successful products received good reviews, 60% of moderately " " " " 10% of poor products received good reviews. In addn, 40% of pdts have been highly successful, 35% have been moderately successful and 25% have been poor products.

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a) What is the prob. that a product attains a good review?
b) If a new design attains a good review, what is the prob. that it will be a highly successful product?
c) If a product does not attain a good review, what is the prob that it will be a highly successful product?

Let G: good review. G' = not good review.
A_1: highly successful
A_2: Moderately successful.
A_3: poor ones.

a) $P(G) = \sum_{j=1}^{3} P(G|A_j)\, P(A_j)$

$= 0.95 * 0.4 + 0.6 * 0.35 + 0.1 * 0.25$

$P(G) = 0.6150$

b) $P(A_1|G) = \frac{P(G/A_1)P(A_1)}{P(G)} = \frac{0.95 * 0.4}{0.6150} = 0.6179$

c) $P(A_1|G') = \frac{P(G'/A_1)\,P(A_1)}{P(G')} = \frac{0.05 * 0.4}{(1-0.6150)} = 0.052$

41 122. An inspector working for a manfr. company has a 99% chance of correctly identifying defective item & a 0.5% chance of incorrectly classifying a good item as defective. The company has evidence that its line produces 0.9% of nonconforming items.
a) What is the prob that an item selected for inspection is classified as defective?
b) If an item is selected at random & classified as nondefective, what is prob that it is good?

A → correctly identifying defective item = 99%
B → Good item as defective – 0.5%.
C → nonconforming items 0.9%. $P(C') = 0$
D → conforming items – 99.1%.
I → chosen item is defective

$P(I) = P(I/C)\,P(C) + P(I/C')\,P(C')$

=

Let D = a defective item is chosen.
C = identifying a defective item

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$P(C) = 0.99$ $P(C') = 0.01$ DATE

$P(C|D)$

I → item is inspected and found defective.
D → item is defective D' = not defective

$P(I|D) = 0.99$ $P(I|D') = 0.005$ % $P(D) = 0.09$

(a) $P(I) = P(I|D)\,P(D) + P(I|D') * P(D')$.

$= 0.99 * 0.09 + 0.005 * 0.991$

$P(I) = 0.094$ 0.0138

(b) $P(D'|I') = \dfrac{P(I'|D') * P(D')}{P(I')} = \dfrac{(1-0.005)(0.991)}{1-0.094}$.

$= 0.999$.

42. Suppose it is known that 1% of the popln. suffers from a particular disease. A blood test has a 97% chance of identifying the disease for diseased individuals, but also has a 6% chance of falsely indicating that a healthy person has the disease.

(i) What is the probability that a person will have a +ve blood test?

(ii) If your blood test is +ve, what is the chance that you have the disease?

(iii) If your blood test is -ve, what is the chance that you do not have the disease?

$P(D') = 0.99$
$P(D) = 0.01$ $P(I|D) = 0.97$ $P(I|D') = 0.06$.

Let D = person having disease.
I = Identification of disease (Testing +ve)

(i) $P(I) = P(I \cap D) + P(I \cap D')$.
$= P(I|D) * P(D) + P(I|D')\,P(D')$
$= 0.97 * 0.01 + 0.06 * 0.99$
$= 0.0691$

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(ii) $P(D/I) = \frac{P(I \cap D)}{P(I)} = \frac{P(I/D) * P(D)}{P(I)} = \frac{0.97 * 0.01}{0.0691}$

$= 0.1404$

(iii) $P(D'/I') = \frac{P(D' \cap I')}{P(I')} = \frac{P(I'/D')\,P(D')}{P(I')}$

$= \frac{[1 - P(I/D')]\,P(D')}{P(I')} = \frac{(1-0.06)(1-0.01)}{(1-0.0691)}$

$= \frac{.94 * 0.99}{0.9309} = 0.9997$

43. Software to detect fraud in consumer phone cards tracks the no of metropolitan areas where calls originate each day. It is found that 1% of the legitimate users originate calls from 2 or more metropolitan areas in a single day. However 30% of fraudulent users originate calls from 2 or more metropolitan areas in a single day. The proportion of fraudulent users is 0.01%. If the same user originates calls from 2 or more metropolitan areas in a single day, what is the prob that user is fraudulent?

S = calls originated from $\geq$ 2 areas.

G = caller being legitimate.

Given $P(S/G) = 0.01$

$P(S/G') = 0.3 \quad P(G') = 0.01$

$P(G'/S) = \frac{P(S/G') * P(G')}{P(S/G') \cdot P(G') + P(S/G)\,P(G)}$

$= \frac{0.3 * 0.01}{(0.3*0.01) + (0.01)(1-0.01)} = \frac{0.003}{0.003 + 0.0099}$

$= 0.2326$

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44. A m/c tool is idle 15% of the time. You request immedi use of the tool on 5 diff occassions during the yr. Assum that your request represent independent events

a) What is the prob that the tool is idle at the time of all of your requests?

b) What is the prob that the m/c is idle at the time of exactly 4 of your requests?

c) What is the prob. that the m/c is idle at the time of at least 3 of your requests?

Soln. a) $(0.15)^5 = 7.6 * 10^{-5}$

b) $[0.15^4 * 0.85) * 5$ 0.00215156

c) P(tool was idle atleast 3 time) =

P(exactly 3 times) + P(4 times) + P(5 times)

$= 10 * 0.15^3 * 0.85^2 + 5 * 0.15^4 * 0.85 + (0.15)^5$

2-146

45. Transactions to a computer database are either new items or changes to previous items. The addition of an item can be completed in less than 100 milliseconds 90% of the time, but only 20% of changes to a previous item can be completed in less than this time. If 30% of transactions are changes, what is the prob that a transaction can be completed in <100 msec?

$(0.2 * 0.3) + (0.7 * 0.9) = \underline{0.69}$

2-151

46. In an automated filling opn, the prob. of an incorrect fill when the process is operated at a low speed is 0.001. When the process is operated at high speed, the probability of an incorrect fill is 0.01. Assume that 30% of the containers are filled when the process is operated at a high speed & remainder are filled when the pro. is operated at a low speed.

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(a) What is the prob. of an incorrectly filled containes?
(b) If an incorrectly filled containers is found, what is the prob. that it was filled during high speed opn?

Let D denote the event that a containes is incorrectly filled & let H denote the event that a containes is filled under high-speed operation. Then,

(a) $P(D) = P(D|H)\,P(H) + P(D|H')\,P(H')$
$= 0.01 * 0.30 + 0.001 * 0.70 = 0.0037.$

(b) $P(H|D) = \frac{P(D|H) * P(H)}{P(D)} = \frac{0.01 * 0.30}{0.0037} = 0.8108.$

47.
2-155. An email message can travel through one of 2 server routes. The prob. of transmission error in each of the servers and the proportion of messages that travel each route are shown in the foll. table. Assume that the servers are independent.

	% of msgs.	probability of errors. server 1	2	3	4
route 1	30	0.01	0.015		
route 2	70			0.02	0.003

(a) What is the prob. that a msg will arrive without error?
(b) If a msg arrives in error, what is prob that it was sent through route 1.

Let E represents Error in msg

$P(E') = (0.3)(0.99)(0.985) + (0.7)(0.98)(0.997) = 0.9764.$

$P(\text{route1}/E) = \frac{P(E/\text{route 1})\,P(\text{route1})}{P(E)} = \frac{(0.3)}{1-0.9764}$

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